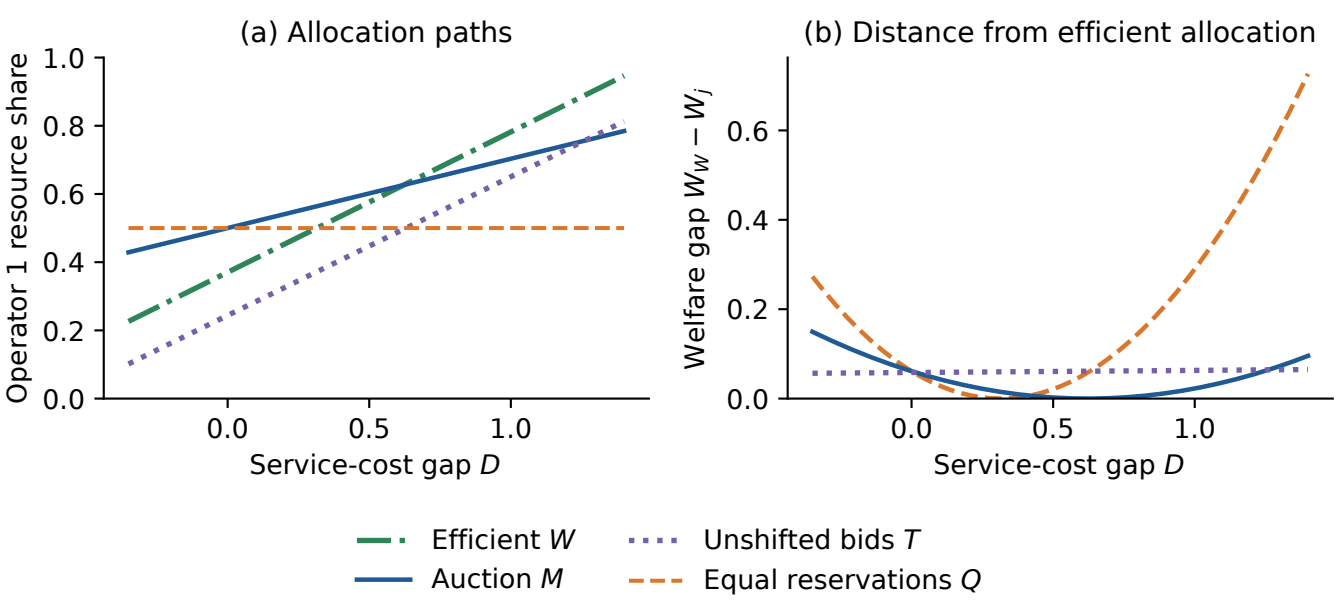
(a) Allocation paths
Operator 1 resource share
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.5
1.0
Service-cost gap D
(b) Distance from efficient allocation
Welfare gap $W_W - W_j$
0.6
0.4
0.2
0.0
0.0
0.5
1.0
Service-cost gap D
Efficient W
Auction M
Unshifted bids T
Equal reservations Q